

Exponential-gap localisation

Claude, with Daniel Murfet

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Abstract

The analytic content of the paper's localisation step (remark `rem:cutoff_correct`): away from the zero set of the phase, where $f \geq \varepsilon$, a Laplace-type integral with amplitude dominated by an integrable g satisfies $\|\int_S a e^{-\beta N f}\| \leq e^{-\beta N \varepsilon} \int_S g$ (`norm_setIntegral_exp_gap_le`); uniformly over dominated families this is $O(e^{-\beta N \varepsilon})$ (`exp_gap_isBig0`), which is o of every power-log scale $N^{-\lambda}(\log N)^r$ (`exp_gap_isLittle0_powLog`). Establishing the gap on the complement of a neighbourhood of the zero set is a separate geometric task. Zero `sorry/axiom`.

1 The things formalised

```
theorem norm_setIntegral_exp_gap_le (S) (hS) (a g f : X → ℝ) (N : ℕ) (hN : 0 < N)
  (hg : IntegrableOn g S) (hbound : x ∈ S, ‖a x‖ ≤ g x) (hgap : x ∈ S, f x ≥ ε) :
  ‖∫ x in S, a x * exp (-( * N * f x))‖ ≤ exp (-( * N * ε)) * ∫ x in S, g x
theorem exp_gap_isBig0 (S) (hS) (a : ℝ → X → ℝ) (g f) (N : ℕ) (h : 0 < N) (hg)
  (hbound : N, x ∈ S, ‖a N x‖ ≤ g x) (hgap) :
  (fun N => x ∈ S, a N x * exp (-( * N * f x)) = 0[atTop] fun N => exp (-( * N * ε)))
theorem exp_gap_isLittle0_powLog (C l) (r) (h : 0 < C) (h : 0 < l) (hC : 0 < C) :
  (fun N => exp (-( * N * ε))) = o[atTop] powLog C l r
```

2 Paper-facing meaning

Together with the normal-block and tangential results, this supplies the third analytic ingredient of the paper's per-stratum expansion: contributions from regions where the phase is bounded below are negligible against every power-log term. What remains geometric is the existence of the gap on $U \setminus U_\varepsilon$ and the chart/partition-of-unity bookkeeping.

3 Lean notes

- `norm_integral_le_of_norm_le` needs only integrability of the dominating function; no measurability of the amplitude or phase is required.
- Beware `mul_comm` rewrites hitting the product inside the exponent first; conclude with `.trans (le_of_eq (mul_comm _ _))` instead.
- `isLittle0_exp_neg_mul_rpow_atTop` gives $e^{-aN} = o(N^b)$; the log power is absorbed by $N^{-\lambda} = O(N^{-\lambda}(\log N)^r)$ for $N \geq e$.